Global Reporting Regulation, Stock Liquidity, and Policy Diffusion

Abstract

We study stock liquidity around global reporting regulation and the international diffusion of those rules. We distinguish 17 economic regulation families and estimate family-specific cohort-time differences before aggregating across families. The first-year comparison covers 15 families. With GDP weights within families and equal weights across families, log spreads and price impact decline by about 0.050. Market-capitalisation weights across families instead yield a log-spread estimate of 0.006. A balanced three-year comparison covers ten families, exhibits adverse pretrends and does not show a broad liquidity improvement. A separate matched-bank analysis finds lower spreads but higher zero-return fractions around the European reporting wave. Independent implementation models do not provide multiplicity-adjusted evidence of diffusion mechanisms. The findings identify differences across policy exposures and weighting targets; they do not establish a general causal liquidity benefit.

1 Introduction

Investors face reporting requirements that differ in their information content, audience and covered population. Comprehensive sustainability reports, bank tax disclosures, confidential multinational tax returns and digital financial filings can therefore affect the information environment through different channels. We ask whether liquidity changes around these rules and whether earlier foreign implementation helps explain subsequent domestic policy choices.

Following the economic sequence in Krueger et al. (2023), we first define reporting obligations, establish the market sample and examine complementary liquidity outcomes. We then use the competition, learning, emulation and coercion framework in Koning et al. (2018) to study implementation. The two parts answer different questions: a market association around an EU rule is not evidence that national policymakers independently learned from foreign benefits.

Our market panel contains 1,055,142 security-years, 99,100 quote identifiers and 90 jurisdictions during 2000–2025. We estimate a separate staggered cohort-time comparison for each economic regulation family and give both equal and market-capitalisation weights across supported families. Banking enters once. Its initial and full reporting phases are not counted as separate equally weighted policies.

The central empirical distinction is the weighting target. First-year equal-family estimates indicate lower spreads and price impact, whereas market-capitalisation-weighted estimates do not establish a broad improvement. The balanced longer-window comparison rejects several pretrend diagnostics. These results favour a differentiated account of reporting environments over an unrestricted average benefit. The extension beyond ESG is the classification and explicit aggregation of heterogeneous obligations; the broader country-market analysis does not reproduce an eligible-firm design.

2 Institutional Setting and Predictions

Public reporting can reduce adverse selection when it supplies credible facts that investors otherwise obtain unevenly. Repeated or immaterial disclosure may add little information, while compliance costs, enforcement differences and concurrent reforms can offset benefits. Narrower spreads and lower price impact are predicted when incremental public information matters. A higher zero-return fraction generally indicates less active price adjustment.

Comprehensive mandatory E/S/G reporting requires a response on all three topics; the last topic determines the joint date. Comply-or-explain still requires a response. CSR spending or a climate-only requirement does not establish comprehensive E/S/G coverage. We retain 37 qualifying reviewed phases from 41 country reviews; earliest separate national topic histories are not complete. Krueger et al. use voluntary earnings guidance as an information-environment proxy, not voluntary ESG disclosure.

Public and confidential CbCR have different audiences. Banking Article 89 reporting initially makes activities, locations, turnover and employment public in 2014; the full annual phase adds profit, taxes and subsidies from 2015. The initial 2014 profit-and-tax submission by globally systemic institutions was confidential. BEPS Action 13 reports go to tax authorities, while general public income-tax reporting concerns a separate multinational regime. Banking coverage does not use the general EUR 750 million threshold.

Eligibility attaches to institutions and their consolidation perimeters. A listed holding company is not automatically the regulated bank. The June 2013 EBA register distinguishes CRD credit institutions from EEA branches and expressly recommends national-register checks. We use it to corroborate bank identities and retain parent and reorganisation gaps. Table 1 lists only implemented phases through 2025, with separate estimation support; it contains no future schedules.

3 Research Design

Annual outcomes are the log median relative quoted spread, log median Amihud price impact and the fraction of observed valid trading-day returns that equal zero. Major primary home equities receive duplicate, delisting, stale-price and extreme-return screens; each outcome requires at least 120 valid observations. The two logged proxies use annual 1/99 winsorisation. The calendar-year and observed-return denominator differ from the reference paper's fiscal-year and potential-day definitions. At least 30 valid securities are required for each country-year outcome.

For family f, cohort g and year t, we subtract the weighted mean change from g-1 among known later implementers from the change among countries implementing in g. Controls remain untreated through the endpoint; unknown histories never become never-treated controls. GDP in g-1 supplies the principal within-family weights. Equal-country and sampled-market-cap weights are disclosed alternatives. Cohort weights equal their eligible treated economic mass and remain fixed across event time.

We report two coequal coverage targets. The inclusive first-year contrast uses g-1 and g, supporting 15 families. The balanced comparison fixes countries and cohorts over g-3 through g+2 and averages years 0–2, supporting ten. The wider endpoint was added to address panel-boundary exclusions before inspecting its coefficients; its design amendment is retained. Differences between the two targets combine horizon and sample changes and cannot be read as a single dynamic path.

Across families, equal weights assign 1/F to each supported family. Market-capitalisation weights use each family's summed sampled-quote capitalisation in g-1 across supported treated cohorts. Both use the same family set across the three core outcomes. The same market can contribute to several regulations: these are policy-exposure weights, not a unique-issuer investment portfolio. Unsupported families are listed rather than entered as zero.

Shared country influence functions and 1,999 common Rademacher draws preserve covariance from countries appearing in several cohorts and families. Intervals are conditional country-sampling diagnostics, not independent-policy randomisation inference. We report joint lead tests, simultaneous event bands, an EU-as-one-block diagnostic and Holm adjustment across 12 coequal principal coefficients. Parallel trends, no anticipation, no interference and absence of unaccounted concurrent reforms remain substantive requirements. Timing shifts, fixed-security samples, independent-choice samples and concurrent-policy exclusions are retained.

4 Liquidity Results

Table 2 presents both requested cross-family weightings. In the inclusive first-year comparison, equal-family log-spread and log-impact estimates are -0.050 and -0.050, approximately 4.9% relative declines. The zero-return fraction declines by 0.00575, or 0.575 percentage points. With market-capitalisation weights across the same 15 families, the corresponding estimates are 0.006,-0.027 and -0.00333. Their conditional intervals include zero. Confidential tax CbCR receives 35.7% of first-year market-cap weight; weighting changes the policy exposure being summarised.

Figure 1 displays every attempted family, including the two unsupported classes. The balanced years 0–2 estimates cover ten families: equal-family log spread is -0.025 and log impact is 0.133; market-capitalisation-weighted values are 0.017 and 0.087. Figure 2 reports the common-support event profiles. Joint lead diagnostics reject for both log-spread aggregates and several other outcomes. The positive balanced price-impact estimate and the pretrends prevent a causal generalisation from the favourable first-year equal-family signs.

Table 3 Panel A examines matched institutions rather than country-market averages. We corroborate 24 EU institution quotes and 21 comparison-group quotes; complete 2011–2016 outcomes leave 22 treated banks in nine home countries and 20 comparison banks in five. Initial-phase mean years 0–2 log-spread contrasts are -0.712 with equal-issuer weights,-0.092 with country-GDP mass allocated across sampled banks, and -0.884 with issuer market-cap weights. Price impact declines, but zero-return fractions rise. These are one-common-wave comparisons subject to pretrends, European banking reforms and possible exposure of comparison groups through EU subsidiaries.

Historical report or exchange-linked identities corroborate three treated banks. The other 21 institution links combine current direct LEI/ISIN evidence with historical quote intervals and the 2013 bank register; they do not certify every historical legal relationship. We report the three-bank identity sensitivity separately. Actual publication is established for Deutsche Bank 2014 on 20 March 2015, Aareal 2014 on 27 March 2015 and HSBC 2014 on 30 June 2015; HSBC's 2013 partial report was published 30 June 2014. Nordea's full 2014 report was published 12 February 2015, but its home-equity identifier remains unresolved. The matched observed-date sensitivity has three publishers, including HSBC's listed parent, and is not a complete staggered filing-date panel.

5 International Diffusion

Following Koning et al., competition uses earlier implementation among export-pattern peers; learning uses prior outcomes among geographic neighbours; emulation uses language, legal-origin or IGO peers; and external dependence uses IMF-credit and aid measures. Every signal precedes the focal year. Family-year strata and country-clustered covariance accompany timing logit and status-model alternatives; controlled specifications include lagged income, growth and trade. The incomplete ledger identifies first recorded phases, not certified first-ever adoption.

Table 3 Panel B reports four focal timing associations. Trade-peer exposure changes sign with controls, and the negative external-dependence coefficient opposes a positive coercion prediction. Across 1,330 attempted designs, 433 stable focal coefficients remain; none passes the broad Holm correction. The bank-identity additions create no new independent national policy choices, so the lagged nonbank input is unchanged and the full matrix is refitted. We do not insert a later estimated liquidity benefit into an earlier policy decision.

6 Conclusion

Liquidity associations differ across reporting rules and across the economic weights used to combine them. First-year equal-family estimates are favourable, while market-cap weights and balanced pretrend checks limit a general liquidity interpretation. The matched-bank study supplies issuer evidence with explicit legal-identity and filing-date boundaries. The diffusion tests do not establish the proposed mechanisms. A causal account requires more complete eligible-issuer histories and policy variation that separates reporting from concurrent reforms.

Table 1. Implemented reporting families and first-year support

Economic regulation family	Country phases	Phase years	Treated markets
Audit oversight / rotation	35	2003–2019	4
Expanded auditor reporting	36	2012–2021	19
Beneficial ownership	18	2016–2024	11
Banking public CbCR	28	2014–2014	16
Confidential tax CbCR	120	2016–2025	42
General public tax CbCR	28	2023–2025	1
Climate disclosure	6	2022–2025	2
Sectoral climate reporting	1	2025–2025	Unsupported
Digital assets	31	2018–2025	2
Comprehensive mandatory E/S/G	37	2002–2022	23
Extractive payments	9	2014–2023	5
Corporate governance	12	1993–2015	6
IFRS reporting	36	2005–2012	22
Supply-chain reporting	8	2012–2024	5
Financial sustainability disclosure	6	2024–2025	3
Informed sustainability reporting	1	2024–2024	Unsupported
XBRL / digital filing	4	2007–2011	1

Country phases are documented country-family implementation records through 2025, not independently verified earliest obligations or eligible-issuer counts. Banking is counted once at its initial public phase; full reporting is a sensitivity. Treated markets denote supported inclusive first-year GDP-weighted contrasts with common three-outcome coverage. Climate-sectoral and informed-sustainability classes lack support. Future schedules and the published ESG benchmark are excluded.

Table 2. All-regulation staggered DiD aggregates

Outcome / coverage target	Equal family estimate [95% interval]	Market-cap family estimate [95% interval]	Holm p equal / cap
A. First year ·15 families Log spread	-0.050 [-0.085, -0.016]	0.006 [-0.032, 0.044]	0.022 / 1.000
A. First year ·15 families Log price impact	-0.050 [-0.093, -0.008]	-0.027 [-0.098, 0.044]	0.113 / 1.000
A. First year ·15 families Zero returns	-0.006 [-0.010, -0.001]	-0.003 [-0.007, 0.001]	0.050 / 0.892
B. Years 0–2 ·10 families Log spread	-0.025 [-0.072, 0.022]	0.017 [-0.054, 0.087]	1.000 / 1.000
B. Years 0–2 ·10 families Log price impact	0.133 [0.052, 0.214]	0.087 [-0.045, 0.219]	0.006 / 1.000
B. Years 0–2 ·10 families Zero returns	-0.004 [-0.011, 0.003]	0.002 [-0.015, 0.018]	1.000 / 1.000

GDP weights within families; both across-family schemes use identical family coverage within a target. Brackets are conditional country-sampling diagnostic intervals. Holm p-values adjust 12 principal coefficients; neither small p-values nor these intervals establish exogenous treatment. Zero returns are fractions. First-year and balanced targets have different countries/cohorts/families. Balanced joint lead p-values: spread 0.0005/0.0005; price impact 0.0005/0.0050; zero returns 0.0020/0.2325 (equal/cap). Complete alternative weights, support failures and fixed design matrix are online.

Table 3. Matched banks and independent policy timing

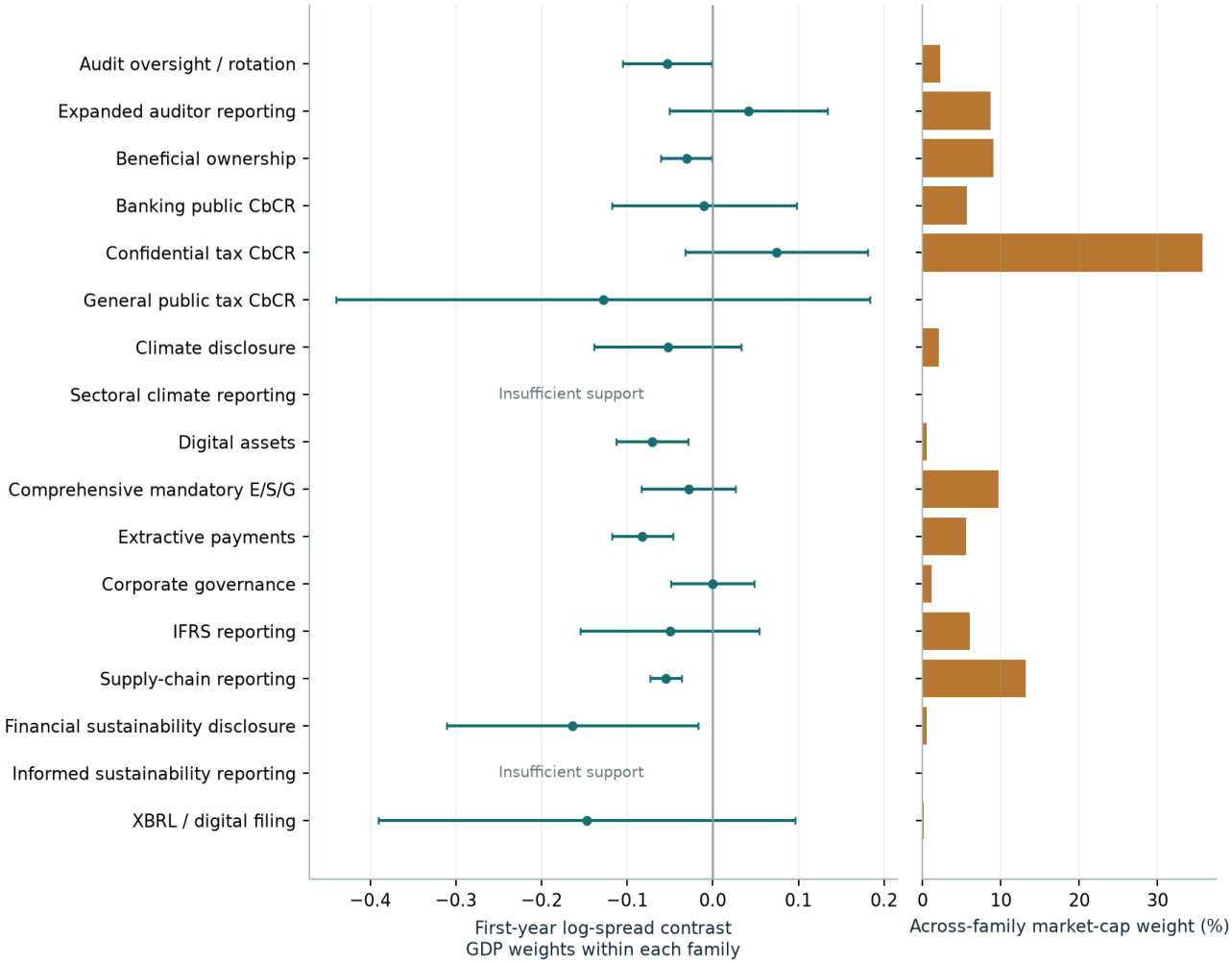
Panel	Weight or mechanism	A: Log spread B: Logit coef.	A: Log impact B: Raw p	A: Zero returns B: Broad Holm p
A. Matched-bank initial phase	Equal issuer	-0.712	-0.246	0.004
A. Matched-bank initial phase	GDP country mass	-0.092	-0.226	0.009
A. Matched-bank initial phase	Issuer market cap	-0.884	-0.121	0.006
B. Independent phase timing	Competition / trade peers	0.458	0.088	1.000
B. Independent phase timing	Learning / GDP growth	-0.409	0.560	1.000
B. Independent phase timing	Emulation / language	0.105	0.703	1.000
B. Independent phase timing	External dependence	-0.720	0.000	0.142

Panel A: 22 treated institutions/9 EU home countries and 20 comparison bank groups/5 nonEU home countries, fixed 2011–2016 support; mean event-years 0–2 against 2013. One collective EU wave; outside-EU groups may have covered subsidiaries. Identity tiers and three-bank dated-report sensitivity are disclosed online. Panel B: pooled documented independent phase-timing logits, one mechanism per uncontrolled model with family-year strata. Continuous focal measures use one sample standard deviation. Broad Holm covers 433 stable focal coefficients across the fixed 1,330-design matrix. Timing associations are not identified diffusion channels.

References

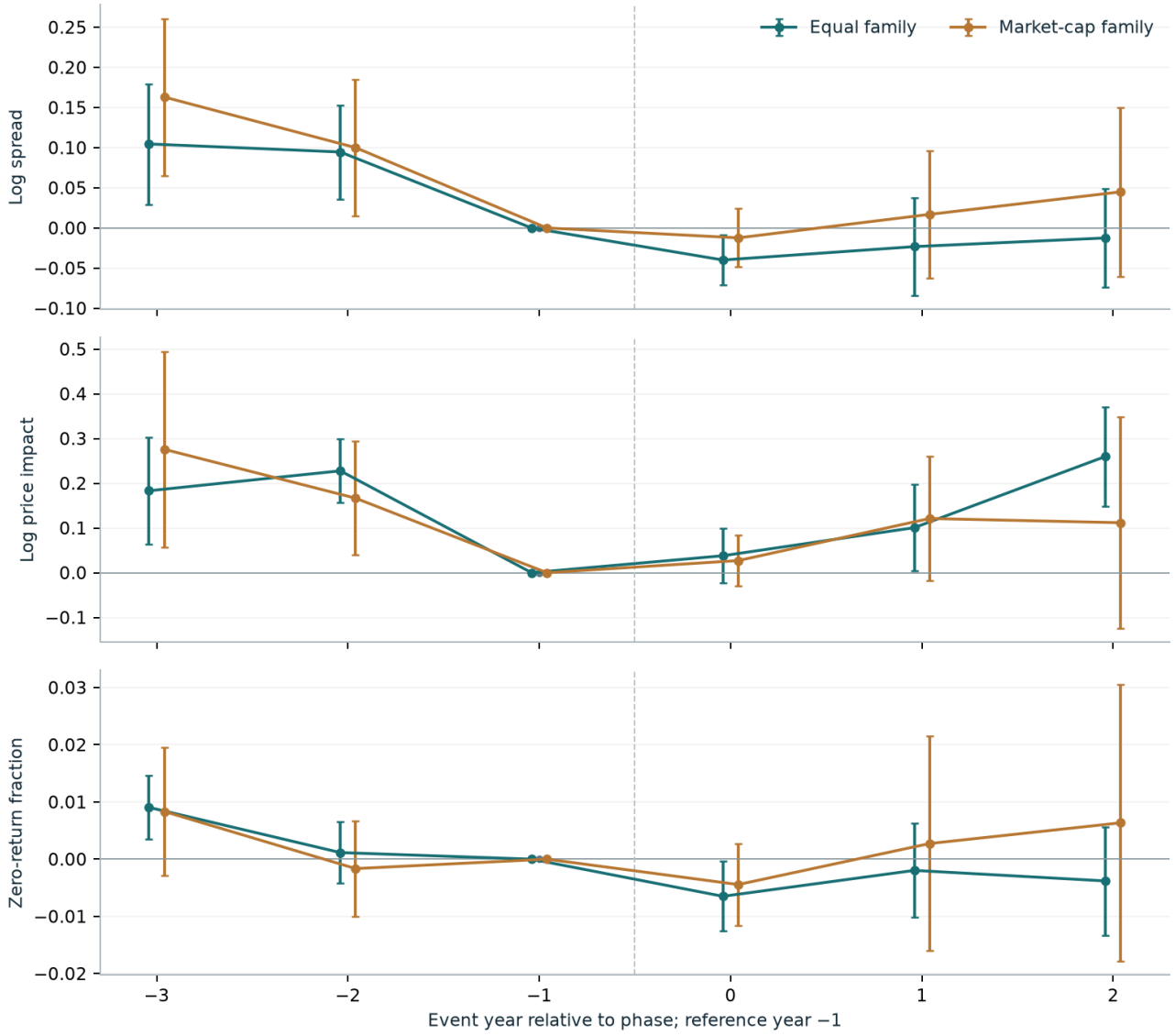
Krueger, P., Z. Sautner, D. Y. Tang, and R. Zhong. The Effects of Mandatory ESG Disclosure Around the World. January 2023 working paper; Journal of Accounting Research 62(2024),1795–1847. DOI:10.1111/1475-679 X.12548.
Koning, M., G. Mertens, and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies 49:249–271. DOI:10.1057/s 41267-017-0123-7.
Callaway, B., and P. H. C. Sant'Anna. 2021. Difference-in-Differences with multiple time periods. Journal of Econometrics 225:200–230.
European Banking Authority. 2013. Register of Credit Institutions, as of 30 June 2013. European Union. Directive 2013/36/EU, Article 89.
Source-linked records and archived originals accompany the online evidence.

Figure 1. Regulation families and their aggregation weights



Inclusive first-year GDP-weighted family contrasts. Bars show conditional 95% country-sampling intervals; right panel shows sampled-quote market-capitalisation weights. Every attempted family is retained. Very small treated-country counts and common EU assignments limit independent-policy inference.

Figure 2. Balanced all-regulation event profiles



Ten-family common-support GDP-weighted comparison, with both requested across-family weights. Reference year-1 is zero by construction. Plotted bars are pointwise 95% conditional country-sampling intervals; simultaneous bands and joint lead diagnostics are in the data. Pre-policy differences are material. This sample differs from the inclusive first-year target.